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Palaeomagnetism of some Late Cretaceous and Miocene igneous rocks on Jamaica

N. H. Guja* and S. A. Vincenz *Department of Earth and Atmospheric Sciences, Saint Louis University, St Louis, Missouri 63156, USA*

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Summary. In order to contribute to the resolution of the problem of the plate tectonic character of the Caribbean, a palaeomagnetic study has been carried out on some Jamaican igneous rocks. Sixteen Late Cretaceous intrusives and lavas and one Late Miocene lava sampled in five sites have been investigated. Because of widespread maghaemitization of the predominantly large-grained deuteric class 1 titanomagnetites, some difficulty was experienced in identifying stable directions of magnetization in the Cretaceous rock units. Using thermal demagnetization technique, two distinct directions of magnetization were obtained, significantly different from those observed in contemporaneous North American rocks. Nine units yield a palaeomagnetic pole at 143.8°W, 44.1°N, referred to as 'normal', while seven units yield 'equatorial' poles situated both east and west of Jamaica. It is not possible to decide which of the two directions of magnetization reflects the Late Cretaceous geomagnetic field in Jamaica, but from other evidence they appear to merit palaeotectonic interpretation. They are consistent with the plate tectonic behaviour of the Caribbean since Late Cretaceous and its motion from the southwest relative to the present day frame of reference, with a simultaneous large anticlockwise horizontal rotation of Jamaica. Late Miocene lavas, containing high-coercivity magnetic material, yield closely grouped directions giving a palaeopole at 152.4°W, 73.3°N, supporting the deductions made from the Cretaceous data.

1 Introduction

Conflicting views exist on how the Caribbean region was formed (see, e.g. Meyerhoff & Meyerhoff 1972; Malfait & Dinkelman 1972). Evidence from focal mechanism studies (Molnar & Sykes 1969; Bracey & Vogt 1970; Isacks & Molnar 1971) points to its plate tectonic character and present-day easterly motion relative to North America. Palaeomagnetic evidence from one or more Caribbean islands may shed some light on its origin. This might be accomplished by obtaining palaeomagnetic samples from one of the greater

* Present address: Institute of Geophysics, University of Teheran, Amirabad, Teheran, Iran.

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Antilles and comparing the results with those obtained from one of the neighbouring continents. Jamaica was chosen for this study. Recently the investigations have in addition been extended to Hispaniola (Vincenz, Dasgupta & Steinhauser 1973a) and corresponding studies of Puerto Rico have been initiated by another research group (Fink & Harrison 1971).

The first palaeomagnetic study of Jamaican rocks was carried out by Watkins & Cambray (1971) on Late Cretaceous dykes. They found that the majority possessed a direction of magnetization indistinguishable from that of the North American Cretaceous rocks and concluded that the palaeomagnetic results did not indicate any rotations of the island.

The present study was undertaken to obtain more data which might yield information of palaeotectonic significance.

2 Geology

Jamaica, situated at approximately 18°N, 77°W is about 250 km long and has a maximum width of 70 km. Under the cover of Tertiary limestones, Cretaceous rocks probably underlie the whole of the island. They are the oldest rocks on the island and appear through the covering in the form of separate sedimentary inliers containing also outcrops of igneous rocks (Fig. 1). Most of the igneous activity occurred in the Late Cretaceous and was accompanied by folding and faulting. It continued into the Palaeocene when it terminated with the injection of the Above Rocks granodiorite pluton.

Widely spread Late Cretaceous rocks occur within the so-called Central Inlier (Fig. 1). Its structure is that of an anticline with a WNW–ESE axis. A region called Wagwater Belt, delineated by two major NW–SE normal faults, was the main scene of Eocene igneous

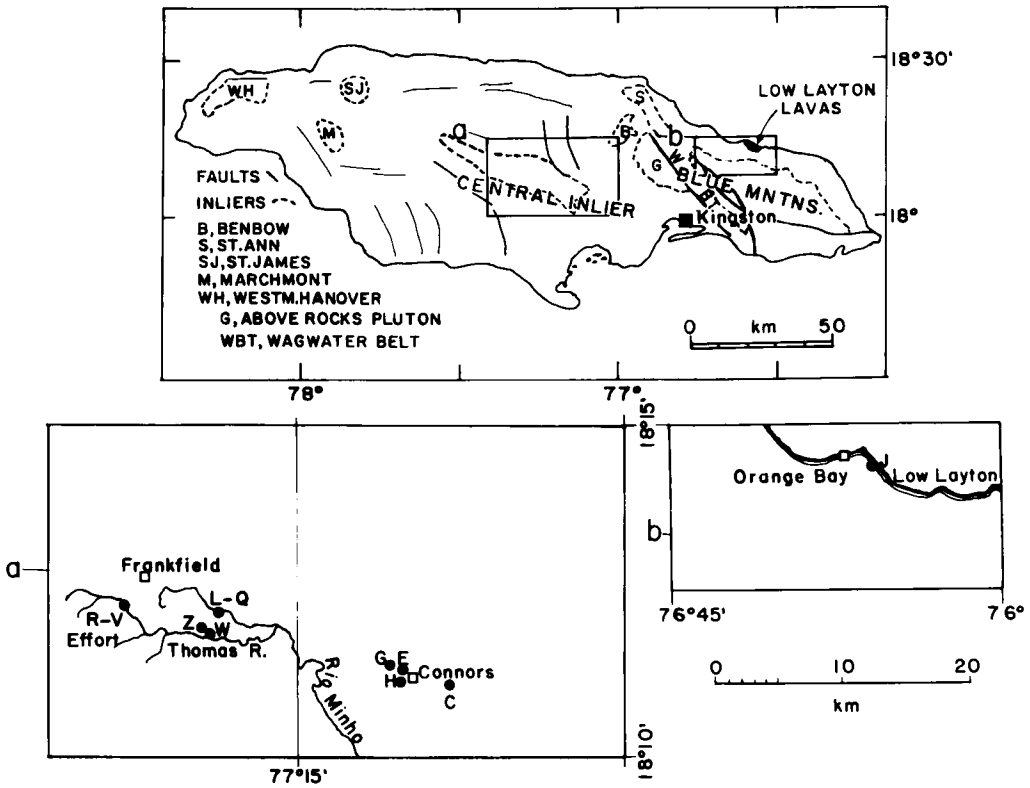


Figure 1. Geological sketch map of Jamaica and location of sampling sites.

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activity and associated strong folding. A period of diastrophism in Mid-Miocene culminated in the extrusion of Late Miocene volcanics of Low Layton near Orange Bay in the Coastal Formation (Fig. 1) on the northeast coast of the island. The generation of the east–west fault system of the island coincides with this period, but maximum movements again occurred in the Wagwater Belt while the Central Inlier was not significantly affected by this activity. As a result of the Late Cenozoic movements, the Late Miocene volcanics were subjected to slight tilting towards the NNE (Zans *et al.* 1962; Reed 1966; Black 1971, private communication; Green 1971, private communication; Roobol 1971).

3 Sampling procedure

After a careful review, a total of 25 igneous units of Late Cretaceous, Eocene and Late Miocene age was selected for sampling. Of these, five Cretaceous and three Eocene units had to be rejected as unsuitable for palaeomagnetic purposes. The remaining 17 Late Cretaceous units and one Late Miocene lava, yielding a total of 248 cylindrical specimens, give significant directions of magnetization. The Late Cretaceous units sampled near the crest of the Central Inlier anticline comprise one granodiorite pluton, one hornblende basalt lava and 15 basalt, andesite or trachy-andesite dykes occurring in several swarms. The units occur in four localities identified as Connors, Rio Minho, Effort and Thomas River (Fig. 1). The Connors group consists of the trachy-andesite dykes C and G, the granodiorite pluton E, and the hornblende basalt lavas H. The Rio Minho dyke swarm comprises dykes L to Q, the Effort swarm dykes R to V, and the Thomas River swarm dykes W and Z. The dykes, striking in various directions, are vertical or near vertical, indicating that their tilting is minimal, as would be expected at the crest of an anticline (Cambray 1969, private communication). This suggests also that the pluton and the lava outcrops have not undergone significant local rotations. Potassium–argon (K–Ar) dates are available only for the Rio Minho dyke swarm (68.7 ± 1.9 Myr: Lewis 1972, private communication) and for the pluton (72.7 ± 2.0 Myr: Reed 1966).

The Low Layton lavas are the only Miocene igneous rocks on Jamaica. The volcanic province, consisting mainly of vesicular pillow lavas, is of a small areal extent (2.5 km long, 1 km wide) and accessible exposures occur only on the sea-shore (Fig. 1). The lavas are intercalated with limestones and palaeontological evidence from the latter clearly points to a Late Miocene age. The regional dip of the coastal formation is 15° to N15° E (Cambray 1969, private communication). Sampling, comprising a total of 95 specimens, was conducted in five sites situated at approximately equal intervals along the accessible strip of the sea-shore about 0.75 km long, with J1 at the western end of the strip and J5 at its eastern end. There was no adequate field evidence to show that each site corresponded to a separate lava flow, but J5, situated somewhat higher than the remaining sites, was in a flow of subaerial origin, while J3 and J4 were in pillows distinct in time from the flows identified by sites J1 and J2. The latter two were situated in fresh rock on the opposite sides of a weathered 'vent' or a feeder 'dyke' striking NNE and about 5 m thick. It is probable that subsidence took place below the 'vent' after the eruption of the lavas and, as subsequent evidence showed, a tectonic correction was required for the directions of natural remanent magnetization (NRM) at the two sites to allow for the rotations caused by subsidence.

4 Experimental results

All measurements were made using astatic and PAR magnetometers with observational error much less than the scatter in directions for intensities greater than 100 nT. The intensities ranged in magnitude from 30×10^3 to 550×10^3 nT.

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Table 1. Typical directional changes on thermal treatment.

Site	*N*	*T*	*D*	*I*	*R*	R_0
P	5	NRM	356.2	46.3	4.86	3.50
	5	100	334.6	44.3	4.73	
	5	200	348.6	40.5	4.82	
	5	300	308.1	31.6	3.97	
S	6	NRM	63.3	51.6	5.68	3.85
	6	100	94.3	39.3	5.37	
	6	200	105.7	18.2	5.72	
	6	300	112.5	9.7	5.60	
	6	400	116.1	9.4	5.85	
	3	500	109.7	8.9	2.83	2.62

Notes:

N is the number of specimens and *T*, the temperature of heating in °C; *D* is the declination, *I*, the inclination, *R*, the magnitude of the sum of *N* unit vectors and R_0, the value of *R* for a random distribution of *N* directions, to be exceeded by observed *R* at the 5 per cent probability level if *R* is significant (Watson 1956; Vincenz & Bruckshaw 1960).

4.1 LATE CRETACEOUS

The results are summarized in Tables 1, 2 and 4 and Figs 2–8. While the NRM directions for the four units of the Connors area differ significantly from the present-day geomagnetic field direction (Fig. 2), those from the Rio Minho dyke swarm show no significant difference, suggesting that the primary remanence is masked by viscous remanence (VRM) of recent origin (Fig. 4). The remaining dykes show mixed directions, some similar to the present geomagnetic field (dykes R and U), others different (dykes S, T and W) and some event at right angles to the magnetic meridian (dykes V and Z) (Figs 5 and 6).

4.1.1 Alternating field demagnetization

To remove secondary components of NRM and study its stability, alternating field (AF) demagnetization was performed. With a few exceptions, the overall picture was one of low stability of the NRM in most of the units under investigation.

Because of the low stability of the primary NRM, AF treatment was generally unsuccessful in producing systematic changes in direction and a significant grouping of directions. In most cases the coercivity ranges of the primary remanence evidently overlapped those of the secondary components which could not be removed except in a few cases, where the primary NRM was stable.

4.1.2 Thermal demagnetization

Thermal demagnetization was found to be a more effective technique for removing secondary components than the AF treatment. Fig. 3 shows the thermal demagnetization curves for the 17 units investigated. Except for lavas H of the Connors group and dykes V and Z of the Effort and Thomas River groups, the thermal stability of the NRM is generally low.

Typical directional changes are illustrated in Table 1 and in Figs 4–6. The final 'clean' directions are summarized in Table 4. The changes were studied by considering mean directions for an average of five to six specimens of each unit.

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Table 2. Opaque mineralogy of Late Cretaceous units.

1	2	3	4	5	6	7	8	9	10
C	Trachytic andesite	a	p		p	1		20–150 (100)	580 (400)
E	Granodiorite (quartz monzonite)	a	p	t		1, 2	s	30–300 (150)	555
G	Andesite	a	t	p		1	l	7–200 (20)	525 (400)
H	Hornblende basalt	a	p		t	1–3	s	5–200 (20)	550
L	Trachyte porphyry	a	t	a		1		10–300 (200)	540 (450)
N	Andesite	a		p		1	m	20–130 (30)	535
O	Andesite	a	t	a		1	l	30–200 (150)	550 (450)
P	Andesite	a		a		1		15–100 (70)	550
Q	Andesite	a		p		1		30–100 (70)	560
R	Andesite	a		p		1	l	20–90 (40)	555 (450)
S	Andesite	a		p		1	m	10–300 (30)	580
Z	Andesite	a	a		t	4–5	m	50–300 (50)	600

Notes:

Column

1. Site.
2. Rock type.
3. Magnetite or titanomagnetite.
4. Haematite or titanohaematite.
5. Maghaemitization of magnetite.
6. Ilmenite.
7. Deuteric oxidation class (Wilson & Watkins 1967; Ade-Hall *et al.* 1968).
8. Hydrothermal alteration: s, strong; m, medium; l, light; blank, none.
9. Grain diameters in μm, dominant size in parentheses.
10. Currie temperature in °C, subsidiary Curie temperature in parentheses. Throughout the table a, indicates abundant; p, present; t, trace; blank, not present.

Although, with increasing temperature, mean directions move away from the direction of the present-day geomagnetic field, in most cases this is accompanied by an increase in scatter, eventually leading to loss of statistical significance. In many cases, nevertheless, the mean direction retains a degree of significance at temperatures up to 500°C as indicated by the observed values of R (Table 1) exceeding its value R_0 for random distribution of vectors (Watson 1956; Vincenz & Bruckshaw 1960). Generally, on heating to temperatures above 300 or 400°C the scatter becomes too large to allow a significance to be attached to the mean direction.

In three Connors units (C, E and H) thermal demagnetization produces no significant directional changes and mean directions remain essentially the same as those of the virgin NRMSs – Figs 3 and 4(a). The results for the six Rio Minho dykes, Fig. 4(b), reveal a systematic movement of mean direction away from the geomagnetic meridian, the most significant change occurring at 300°C. In view of the erratic behaviour of individual specimens above 300°C, heating of specimen batches above this temperature would have served

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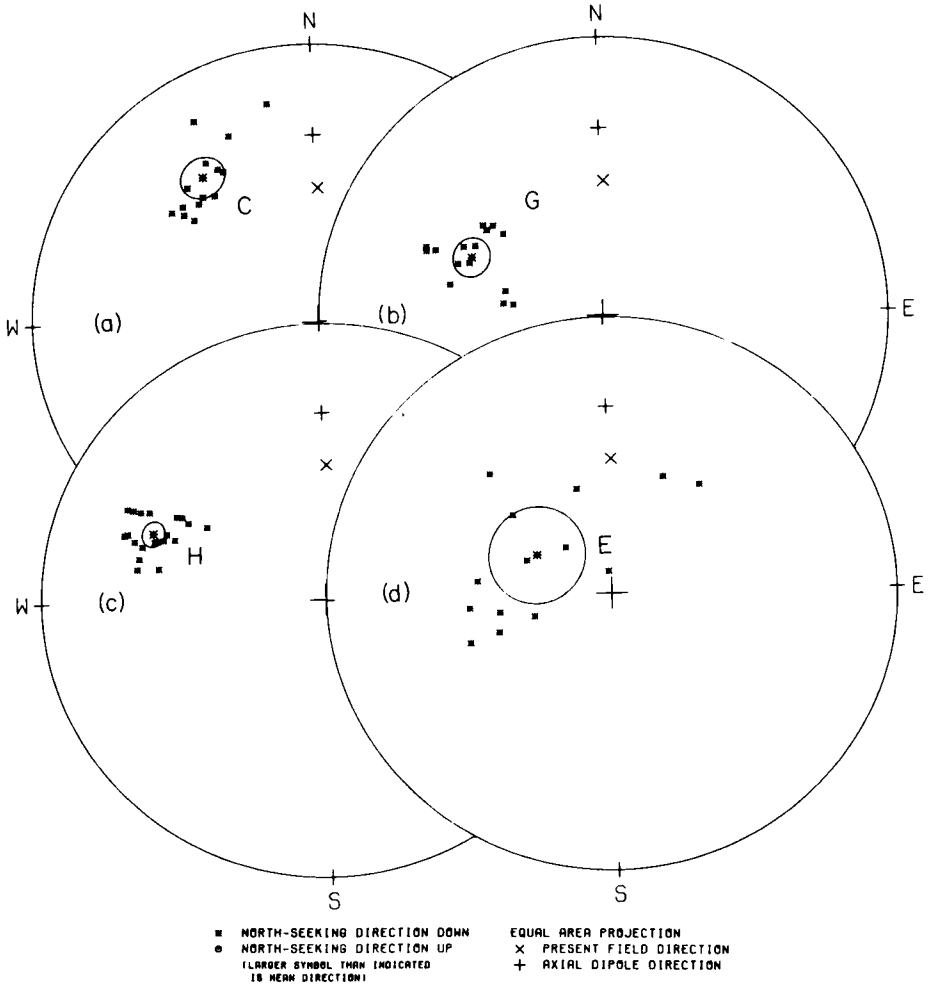


Figure 2. NRM directions of the Connors group; smaller circles are the α_{95} circles of confidence (Fisher 1953).

no useful purpose. The final direction at 300°C occurs close to the mean NRM direction for the Connors group from which dyke G has been excluded because of its behaviour unrelated to the other three Connors units. Lack of significant directional change in these three units suggests that despite the large error circle, Fig. 4(a), the mean virgin NRM direction approximates that of the original cooling thermoremanence (TRM). Since the mean direction of the Rio Minho dykes at 300°C is close to the Connors mean, this supports the above conclusion, but increase of scatter on heating to higher temperatures indicates that it is not certain that a full removal of secondary remanence has been achieved.

The directional changes in Effort dykes are consistent in dykes S, T and V, but not in dykes R and U (Fig. 5). The easterly equatorial direction found in dykes S, T and V is approximately matched by some specimens from dykes U and R, but the majority give a westerly direction and in the case of dyke R a direction close to the mean direction of the Connors group – Fig. 5(c), (e) and (f).

Thermal demagnetization of dykes W and Z shows a change in direction for dyke W, but

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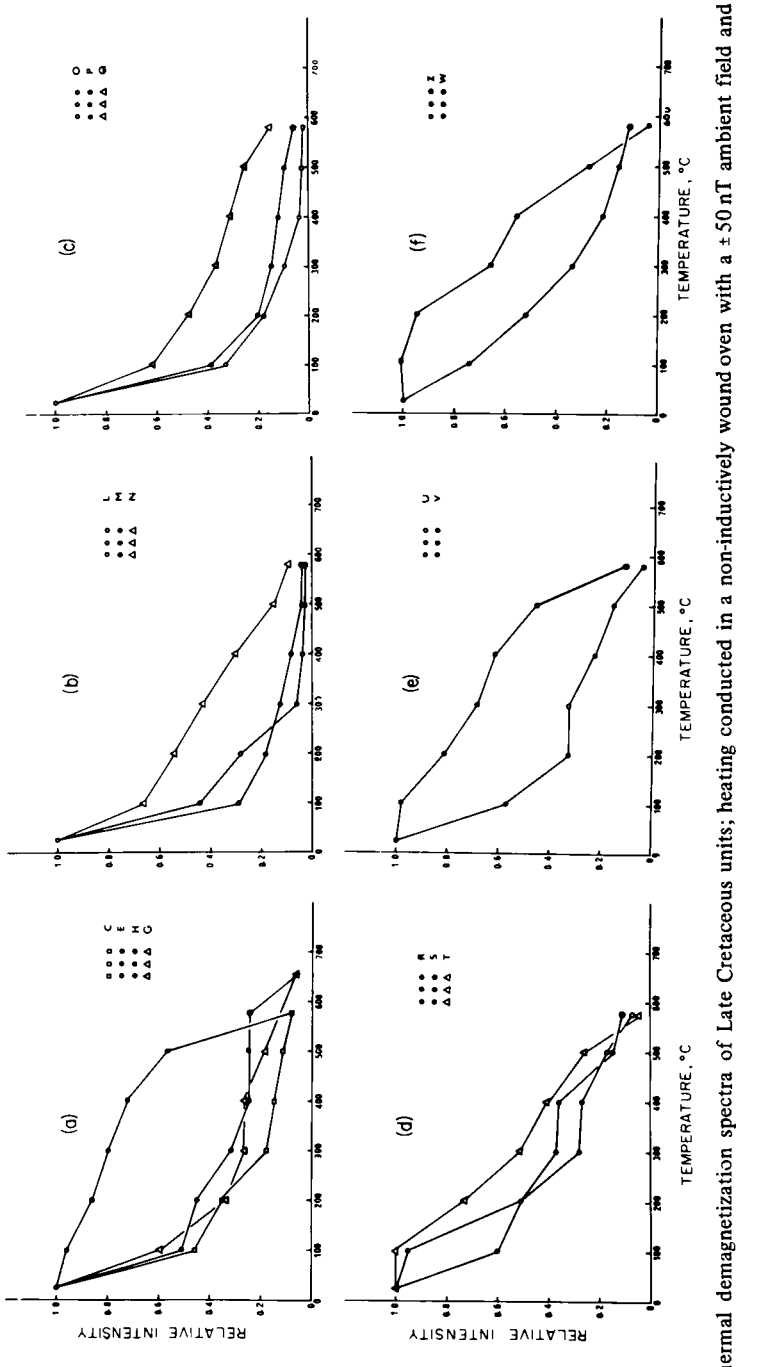


Figure 3. Thermal demagnetization spectra of Late Cretaceous units; heating conducted in a non-inductively wound oven with a ±50 nT ambient field and nitrogen atmosphere.

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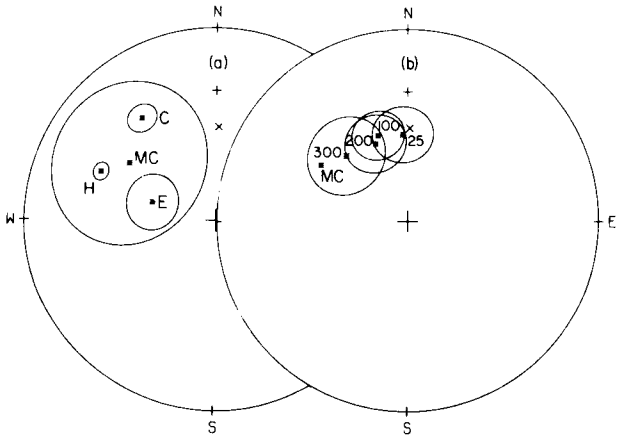


Figure 4. (a) Mean NRM directions of three Connors units; (b) temperature dependence of mean NRM direction of six Rio Minho dykes; MC, mean direction of three Connors units; temperatures of heating are indicated; 25°C corresponds to NRM; other symbols as in Fig. 2.

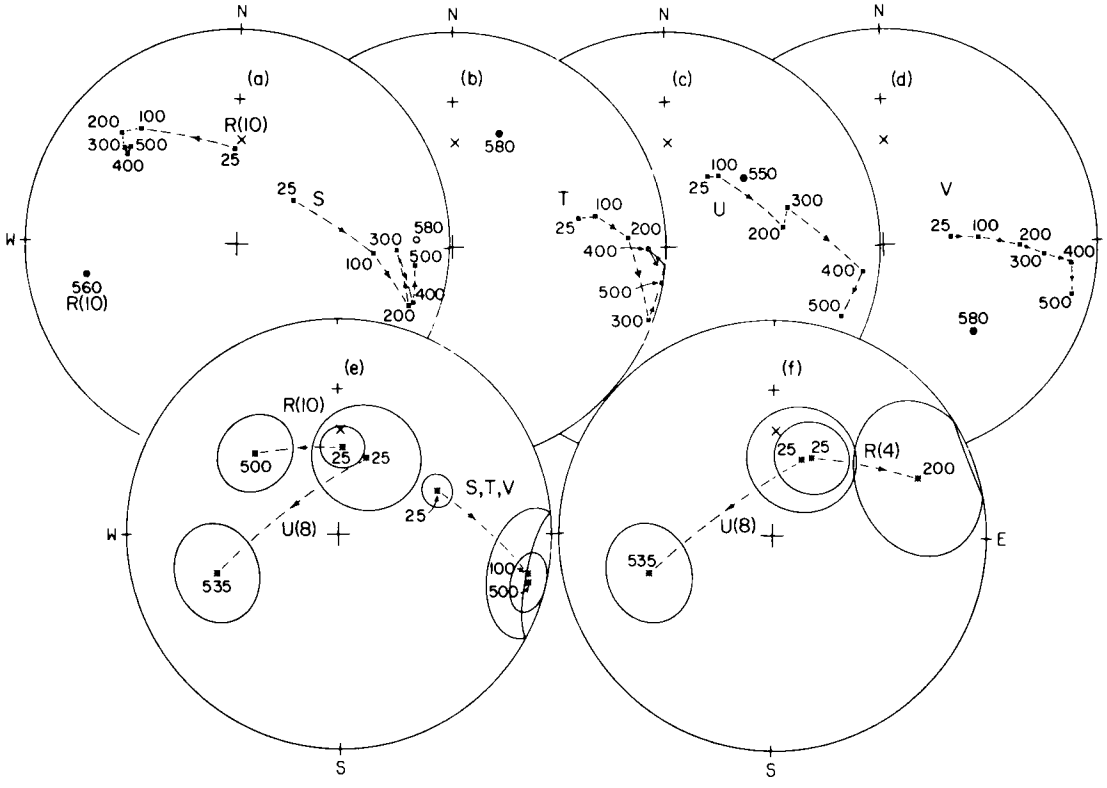


Figure 5. Directional changes of NRM on thermal demagnetization of Effort dykes: (a), (b), (c) and (d), representative individual specimens. Mean directions: (e), dykes S, T and V, eight specimens from dyke U (U(8)) and ten-specimen group from dyke R (R(10)); (f), dyke U(8) and four-specimen group from dyke R (R(4)); hexagonal symbols denote directions obtained at temperatures at which intensity observations became unreliable; legend as in Fig. 2.

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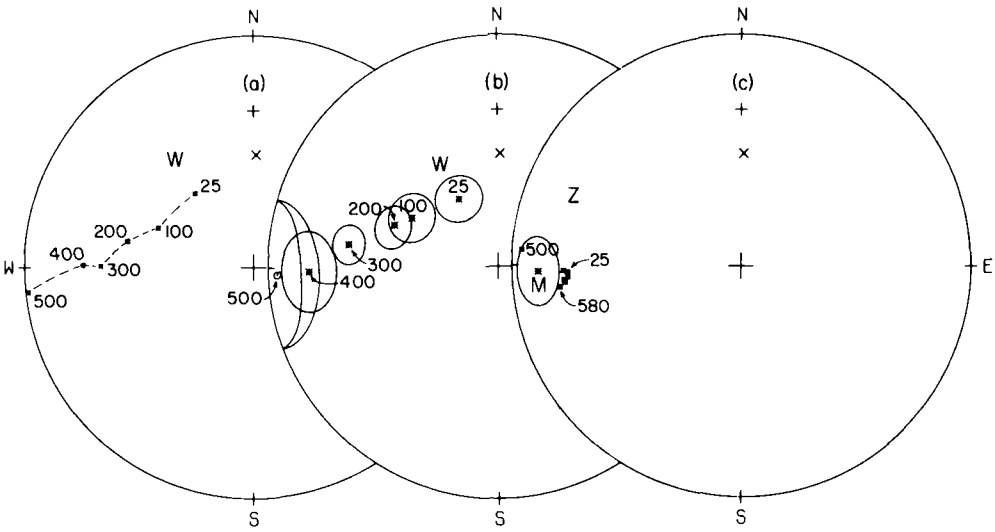


Figure 6. Directional changes of NRM on thermal demagnetization of Thomas River dykes: (a), representative specimen from dyke W; (b), mean direction for seven specimens from dyke W; (c), representative specimen from dyke Z and mean NRM direction of Z; legend as in Fig. 2.

no significant change for dyke Z (Fig. 6). The directions in W move towards the virgin mean of Z, which is approximately antipodal to that observed in dykes S, T and V.

4.1.3 Identification of magnetic minerals

Fig. 7 shows typical thermomagnetic curves and Fig. 8 represents microphotographs of magnetic minerals seen in polished sections. The results of identification studies are summarized in Table 2. With some exceptions the predominant magnetic mineral is class 1 titanomagnetite or magnetite (highest Curie temperatures: 525–578°C) which shows signs of widespread maghaemitization, in some cases the maghaemite having been further altered to haematite. In several cases (dykes C, L, O and G), two titanomagnetite size generations are present. In dyke C some class 1 grains are partly haematized, suggesting the possibility of occurrence of higher deuteric classes. In some instances limited granulation relates to effects of regional hydrothermal alteration (Ade-Hall, Palmer & Hubbard 1971), also indicated by thin-section studies showing plagioclase partly altered to sericite and chlorite. Alteration of this type was observed in dykes N, O, R, S and Z.

The widespread maghaemitization, previously observed by Watkins & Cambray (1971), reflects presumably the effects of surface weathering in a marine tropical climate. An exception to this are the lavas at site H, which reveal no maghaemitization of their class 1 to 3 titanomagnetite. The latter has, however, been reheated above 600°C, as manifested by pseudobrookite replacing the pre-existent ilmenite lamellae – Fig. 8(d). The reheating evidently took place at the time of the emplacement of the lavas, because there is no evidence of any later igneous activity in the immediate vicinity of the lavas.

The magnetic mineralogy of dyke Z is one of a high-grade deuteric oxidation andesite with large class 4–5 grains and all small grains haematized and possibly a regional hydrothermal alteration overprint. Pluton E contains large class 1 or 2 titanomagnetites, all showing a slight tendency towards martitization and/or replacement by haematite aggregates. The haematite fraction is, however, too small to show its effect on the thermomagnetic curve which gives a Curie temperature of 555°C (Fig. 7). Replacement of ilmenite

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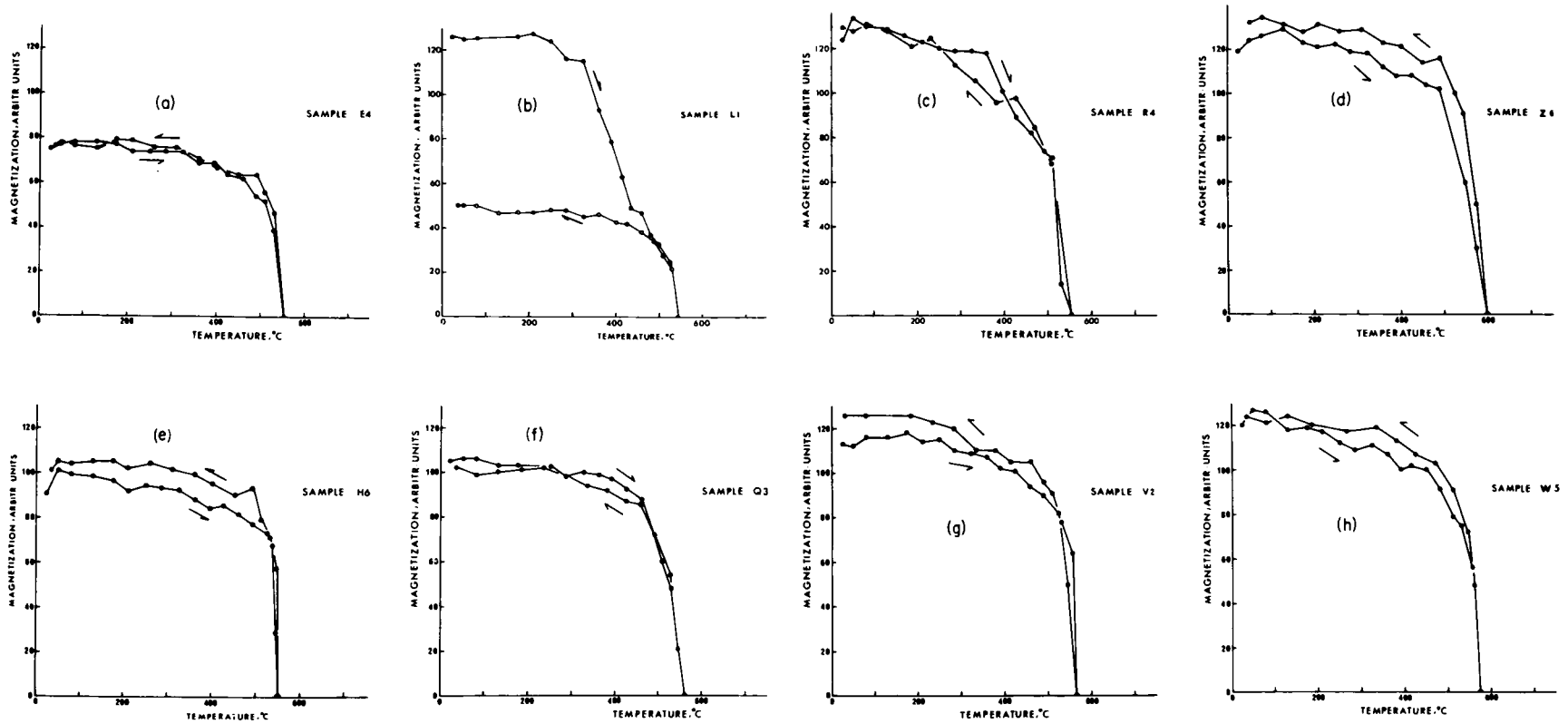


Figure 7. Determination of Curie temperatures of Late Cretaceous units. Solid circles, heating; open circles, cooling. Magnetizing fields: (a), (e) and (f), 60 mT, (b) and (c), 80 mT; (d), 40 mT; (g) and (h), 50 mT. Heating conducted in nominal vacuum (about 10^{-3} T).

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Figure 8. Photomicrographs of titanomagnetites of Late Cretaceous units, taken using a Zeiss WL-POL reflection microscope and oil immersion. Connors units: (a) dyke C, partly haematized (some maghaemite present) class 1 titanomagnetite; titanomagnetite, grey; maghaemite, off-white; haematite, white; some tiny sulphide inclusions, bright white; (b) granodiorite E, peripherally martitized class 1 titanomagnetite; (c) dyke G, two-size generations of speckled maghaemitized class 1 titanomagnetite; (d) lavas H, uniform class 2 to 3 titanomagnetite with pseudobrookite replacing the pre-existent ilmenite lamellae and alteration feature near centre, both suggestive of early reheating *in situ* above 600°C. Rio Minho dykes: (e) dyke L, maghaemitized class 1 titanomagnetite; note the annular zoning showing the novel feature of repeated maghaemitization followed by deposition of new titanomagnetite; (f) dyke 0, almost wholly maghaemitized class 1 titanomagnetite. Effort dykes: (g) dyke R, class 1 titanomagnetite with wormy (maghaemite) alteration and a little granulation; (h) dyke S, anhedral class 1 titanomagnetite, patchy light and dark colouration and presence of amorphous iron-titanium oxide suggest effect of regional hydrothermal alteration. (i) Thomas River dyke Z: class 4 to 5 titanomagnetite with a regional hydrothermal alteration overprint. All dimensions are given in μm.

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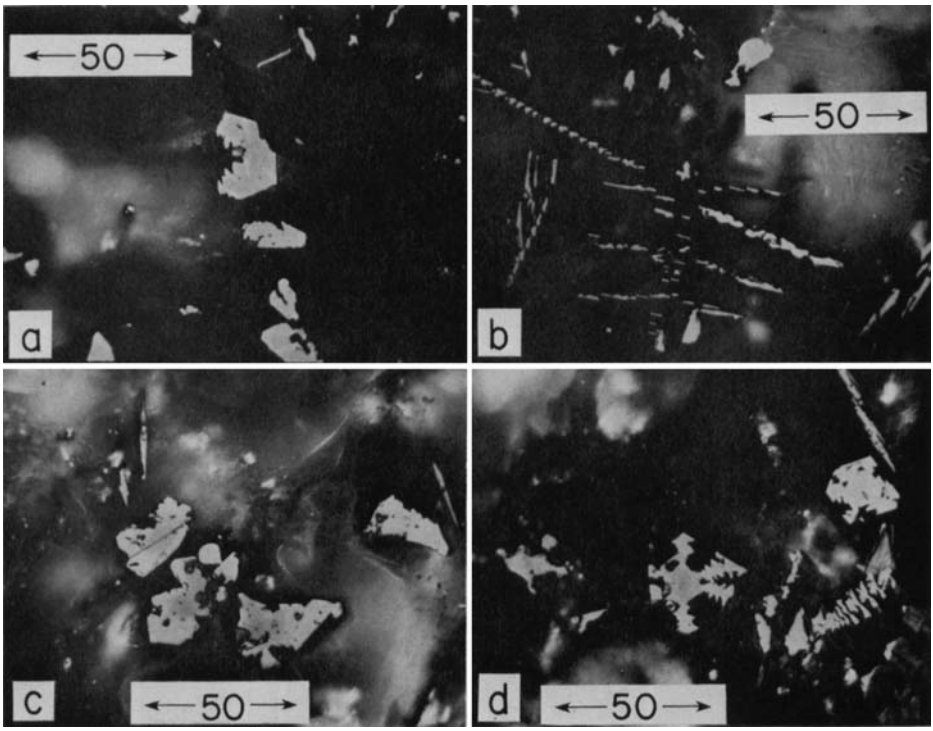


Figure 11. Photomicrographs of magnetic minerals in Low Layton lavas. Site J1: (a) patchilly maghaemitized class 1 titanomagnetite; (b) totally or partly maghaemitized skeletal class 1 titanomagnetite and skeletal acicular ilmenite (dark grey), note the grain of haematite near the centre. (c) Site J3: largely maghaemitized skeletal class 1 titanomagnetite and skeletal to euhedral ilmenite. (d) Site J4: patchilly maghaemitized class 1 titanomagnetite and unaltered acicular ilmeuite. All dimensions in μm.

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lamellae in class 2 grains by sphene and presence of large translucent relics of opaques suggest that the pluton has been subjected to regional hydrothermal alteration at about 300°C. The alteration may have taken place during the initial cooling or has been the result of a post-crystallization process (Lewis 1969, private communication).

Since, with the exception of dyke Z, all the Late Cretaceous units are characterized by low-grade deuteric oxidation, they are expected to have soft remanences, unless the grain sizes are small enough as in lavas H which did not require any thermal 'cleaning'. Widespread presence of maghaemite suggests, however, that in most cases the original cooling TRM is partially masked by secondary remanences of chemical or viscous origin. The presence of maghaemite is illustrated by one thermomagnetic curve, Fig. 7(b), showing that heating in the laboratory has resulted in loss of intensity of magnetization corresponding to conversion of maghaemite into haematite. Absence of this effect in other curves confirms that in many cases maghaemite was already haematized *in situ*, as observed in several polished sections.

4.2 LATE MIOCENE

Fig. 9(a) summarizes the results of NRM measurements. The direction of magnetization is reversed and different from the direction of the present geomagnetic field. Fig. 10 describes the results of thermal and AF demagnetization and of the Curie point determination. The polished section studies are summarized in Fig. 11 and Table 3.

The lavas contain partially to totally maghaemitized euhedral to skeletal class 1 magnetite or titanomagnetite and abundant skeletal to acicular ilmenite. Sites J1 and J2 contain

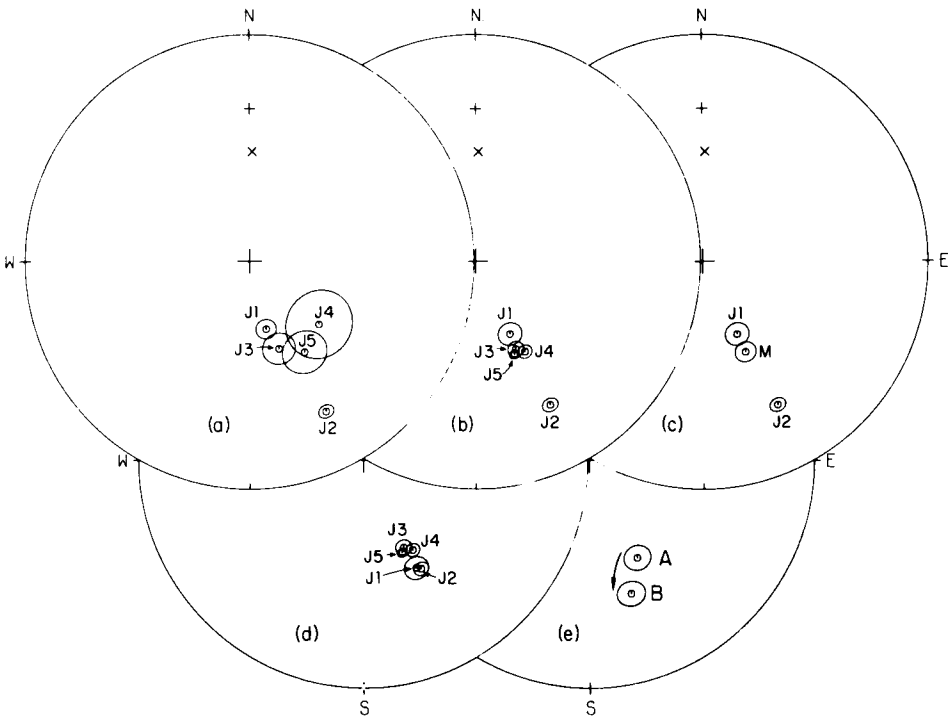


Figure 9. Directions of magnetization of Low Layton lavas: (a) NRM; (b) after selective thermal demagnetization; (c) J1 and J2 as in (b), M, mean direction of J3, J4 and J5 after thermal demagnetization; (d) as in (b) J1 and J2 corrected for subsidence; (e) final mean for five sites before (A) and after (B) tectonic correction. Legend as in Fig. 2.

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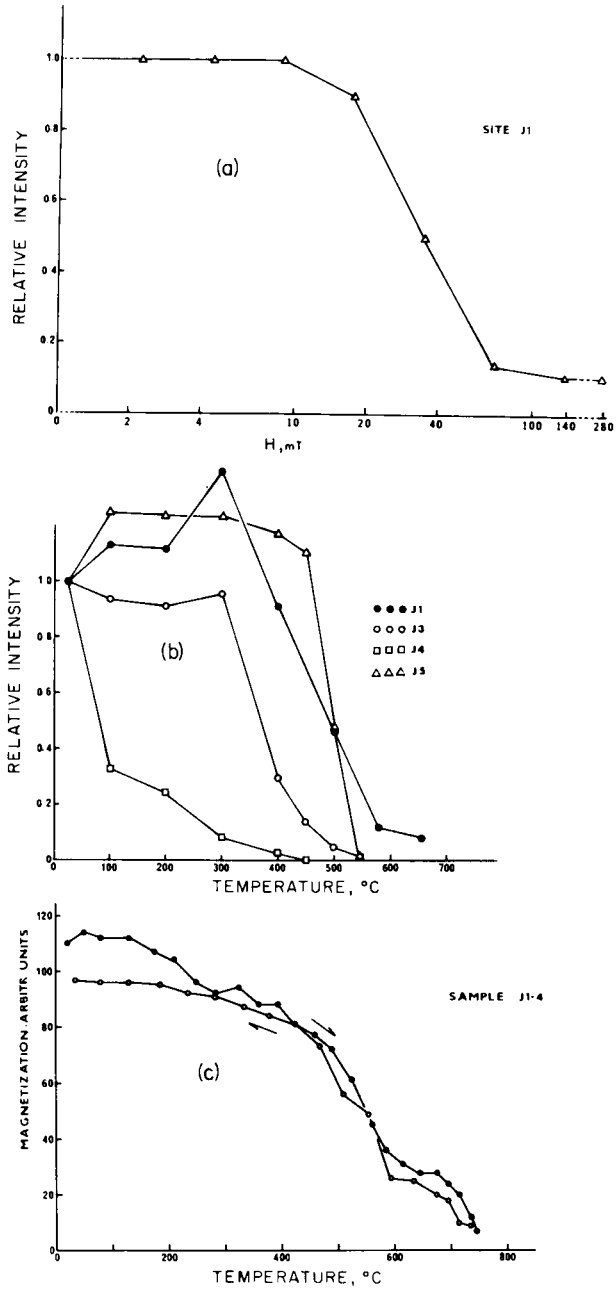


Figure 10. Magnetic characteristics of Low Layton lavas: (a) AF demagnetization; (b) thermal demagnetization; (c) Curie temperature determination, magnetizing field 90 mT.

also some haematite evidenced by Curie temperatures near 700°C. The grain sizes are small, averaging 5–25 μm and suggesting that the NRM should be stable. At site J5 two size generations of titanomagnetite are present. Since the deuteric oxidation index is low, most of the ilmenite is primary. The abundance of maghaemite must be in part a result of weathering which is indicated by the observed reddening of the silicates. Significant difference of the

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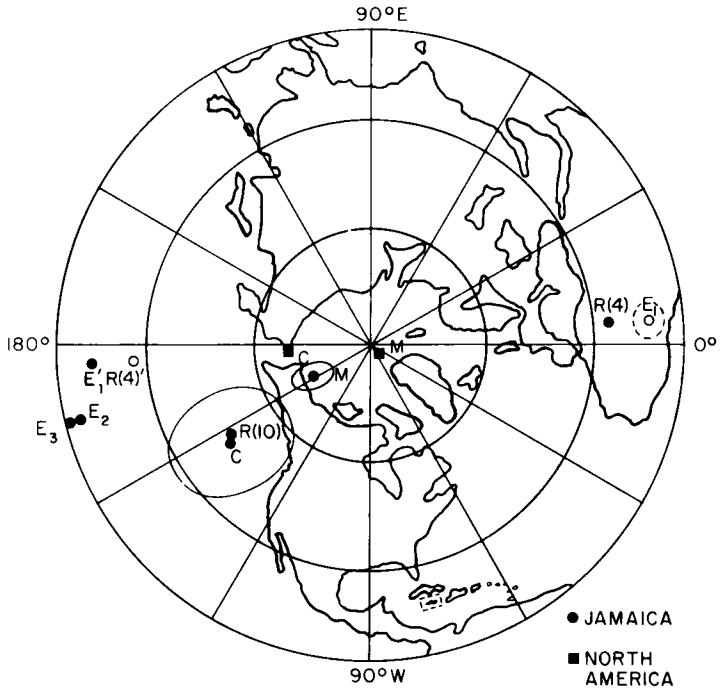


Figure 12. Summary of palaeomagnetic pole positions: C, Cretaceous; M, Miocene; full symbols, northern hemisphere; open symbols, southern hemisphere. C for Jamaica computed using nine 'normal' Late Cretaceous units. 'Equatorial' poles: E_1, mean for dykes S, T and V; E_2, mean for six and E_3, mean for seven dykes listed in Table 4. R(4) and R(10) refer to two distinct groups of specimens from dyke R. Primed symbols denote pole positions with reversed polarity. Ovals of confidence as in Table 4.

NRM direction from that of the present geomagnetic field suggests, however, that most of the maghaemitization resulted from a low-grade post-formational contemporaneous hydrothermal alteration process.

AF demagnetization curve indicates the presence of high coercivity material, but abundance of maghaemite, also evident from the reduction of remanence seen in the cooling part of the thermomagnetic curve, Fig. 10(c), suggests that the removal of secondary remanence should be preferably conducted by thermal demagnetization. Both AF and thermal treatment show an increase of remanence at some stage of the demagnetizing process, Fig. 10(a)

Table 3. Opaque mineralogy of Late Miocene lavas.

1	2	3	4	5	6	7	8	9	10
J1 and J2		a	p	a	a	1	1	1–80 (5)	580 (700)
J3	Andesite porphyries	a	t	a	a	1	1	5–50 (25)	560
J4		a	t	a	a	1	1	5–50 (25)	450
J5		a	t	a	a	1	1	1–30 (30)	550

Note:
Notation as in Table 2.

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Table 4. Summary of 'clean' palaeomagnetic results.

Age and designation	Site or group of sites	N	D	I	k	α_{95}	ϕ	λ	δ_p	δ_m	Polarity
Late Cretaceous 'normal'	C	5	318.8	21.5	10.5	24.6	171	50	14	26	(–)
	E	12	285.1	61.8	15.8	11.2	127	23	14	17	(–)
	H	20	292.3	35.4	89.3	3.5	154	26	2	4	(–)
	L	6	311.2	40.6	9.6	22.7	152	44	18	23	(–)
	M	6	284.1	57.8	39.4	10.8	132	22	13	16	(–)
	N	6	304.5	70.9	17.4	16.5	112	34	25	28	(–)
	O	6	340.6	50.2	22.1	14.6	128	68	14	20	(–)
	P	5	308.1	31.6	3.9	44.6	160	41	29	50	(–)
	Q	6	342.4	48.0	5.6	31.2	130	70	28	42	(–)
	Mean	9	311.0	48.1	15.1	13.7	143.8	44.1	12.3	18.0	(–)
	R(10)	10	313.8	45.0	12.2	14.4	147.6	46.6	11.2	17.4	(–)
	Mean (with R(10))	10	311.4	47.8	16.9	12.1	144.3	44.5	10.3	14.8	(–)
	S	3	109.8	8.9	12.2	37.0	178	17	18	37	(+)
	T	3	102.4	12.3	17.6	30.3	177	10	16	31	(+)
	V	4	100.7	5.9	221.7	6.2	174	9	3	6	(+)
Late Cretaceous 'equatorial'	Mean (S, T, V)	3	104.3	9.1	201.5	8.7	176	12	4	9	(+)
	R(4)	4	68.1	27.8	13.6	25.8	176	25	16	28	(+)
	U	8	252.5	39.6	11.2	17.3	140	8	13	21	(–)
	W	7	268.1	19.1	28.7	11.4	158	2	6	12	(–)
	Z	8	268.6	12.4	36.6	9.3	161	1	5	10	(–)
	Mean (S, T, V, U, W, Z)	6	274.7	7.4	12.0	20.2	165.3	5.6	10.4	20.8	(– +)
	Mean (with R(4))	7	271.0	2.5	9.4	20.8	165.3	1.7	10.3	20.4	(– +)
Late Miocene	Mean	5	162.6	–38.5	269.5	4.7	152.4	73.3	3.4	5.6	(+)

Notes:

N is the number of specimens or sites, k, the estimate of the precision parameter and α_{95}, the radius of the circle of confidence (Fisher 1953). ϕ is the western longitude and λ the northern latitude of the palaeomagnetic poles and δ_p and δ_m the semiaxes of the oval of confidence; R(10) and R(4) refer to the mean directions of two specimens groups from site R, obtained at 500 and 200°C respectively; the polarity is given for the pole position quoted: (–) is the normal polarity, (+) means that the observed direction has been reversed, (– +) refers to the position computed using both normal and reversed directions; for clarity all poles are given in the Pacific hemisphere; other symbols as in Table 1.

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and (b), evidently related to the removal of VRM of recent origin. The contribution of the chemical remanent magnetization (CRM) carried by haematite in sites J1 and J2 is apparently insignificant, suggesting that the haematite is either primary or of post-formational contemporaneous origin. A selective heat treatment up to a maximum of about 450°C – see the curves for J1 and J5, Fig. 10(b) – would thus be expected to remove most of the secondary remanence, and the resulting directional changes, obtained using the minimum scatter principle, are illustrated in Fig. 9(b). These directions are different from the present-day geomagnetic field direction, but there is a significant difference between the mean directions of sites J1 and J2. This difference can be accounted for by the fact that both sites are situated on opposite sides of the feeder 'dyke' described earlier in this paper. Subsidence below the 'dyke' after the cessation of volcanic activity apparently produced rotations in the opposite sense of the lava units on both sides of the 'dyke'. The corresponding remanence vectors were, therefore, also rotated, causing their mean directions to differ significantly from the mean directions of sites J3, J4 and J5 – Fig. 9(b) and (c). Subsidence under volcanic vents after the cessation of activity is a common phenomenon (DeSitter 1956). Using the observed strike direction of the 'dyke' (N 15°E), a correction was applied to the directions of J1 and J2 for the local collapse under the 'dyke'. Each mean direction was rotated clockwise and anticlockwise respectively through an estimated angle of 15° and the resulting corrected directions are shown in Fig. 9(d).

The overall mean for five sites, corrected for the regional dip, is shown in Fig. 9(e). The numerical details of the final result are given in Table 4.

5 Discussion

5.1 LATE CRETACEOUS

A dominant feature of the Jamaican Late Cretaceous igneous units is the widespread maghaemitization attributed to the tropical weathering process. Generation of maghaemite has evidently resulted in imposition of the CRM as well as of VRM, both of which have the direction of the recent geomagnetic field, and when pronounced, dominate the original cooling TRM carried by the residual titanomagnetite (Dasgupta & Vicenz 1975).

The question arises to what extent it has been possible to remove the secondary components so that the remaining NRM largely corresponds to the original cooling TRM. This dilemma is complicated by the fact that in some of the rock units a part of the maghaemite has been converted to haematite carrying a CRM which cannot be removed without destroying the original TRM. Large grain sizes and presence of maghaemite are evidently responsible for the increase in scatter observed on heating above 300°C and related to overlapping blocking temperature spectra of the magnetic constituents. The results show, however, that the mean NRM directions of three Connors units were not materially affected by laboratory heating. This suggests that the reheating *in situ* of two units (E and H), deduced from polished section studies, was post-formational contemporaneous and did not significantly modify the direction of the original cooling TRM. It seems, therefore, probable that the mean virgin NRM directions of the three units are, in fact, those of the original TRM. For purposes of classification we shall refer to these directions, as well as to that obtained after thermal demagnetization of the Rio Minho dykes, Fig. 4(b), as 'normal', while the directions of the Effort and Thomas River dykes will be termed 'equatorial' (Figs 5 and 6).

The case of dyke G is special: its mean direction of magnetization changes systematically with no significant change in scatter from $D = 302.8°$, $I = 45.2°$ at room temperature to $D = 172.6°$, $I = 55.0°$ at 400°C. This systematic change and presence of two size generations of magnetite suggest that this dyke was reheated *in situ* in an ambient field of different

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direction. The observed virgin NRM direction evidently represents a resultant of several components, and the original remanence may have been acquired during a polarity transition of the Late Creteaceous geomagnetic field. Since, in addition, geological field evidence showed that there have been some earth movements in the immediate vicinity of the dyke, it was decided to exclude it from further analysis.

The mean direction of six Rio Minho dykes after heating to 300°C is similar to that of three Connors units (Fig. 4). The Effort dykes also show some maghaemitization, but three of them (S, T and V) reveal a systematic directional change on thermal treatment, giving a mean direction at 500°C approximately at right angles to the present magnetic meridian (Fig. 5(e)). Two types of directions, 'normal' and 'equatorial' are identified in dykes R and U, Fig. 5(a), (c), (e) and (f), which belong to the same swarm as S, T and V and are only a few metres apart. While dyke T cuts dyke S, dykes S and R are parallel and U is approximately parallel to dyke S. Thus the only plausible explanation for the mixed directions of magnetization is that they have resulted from reheating in a field of changed direction, such as might have occurred during a polarity change of the geomagnetic field.

The 'clean' direction of magnetization of the two Thomas River dykes is westerly 'equatorial' and similar to that observed in eight specimens of dyke U.

'Normal' and 'equatorial' remanence directions have been observed in other Jamacian rocks. The results of these investigations have been reported separately and relate to Early Cretaceous lavas (Steinhauser & Vincenz 1973), Late Cretaceous ash flows (Vincenz, Steinhauser & Dasgupta 1973b), and the Above Rocks Palaeocene pluton (Dasgupta & Vincenz 1975).

The question arises which of the two types of directions, the 'normal' or the 'equatorial', represents the original cooling TRM. Except for dyke G, earth movements do not appear to have been a factor likely to have affected the direction of magnetization.

Allowing for the scatter of data, the 'equatorial' directions represented by dykes S, T and V and those of dykes U, W and Z are approximately antipodal and the 'normal' directions lie on a great circle path between the present-day field direction and the 'equatorial' directions. There are several possible explanations of the data. These include effects of polarity transitions of the reversing 'normally' directed Cretaceous geomagnetic field, imperfect removal of secondary components of NRM, contaminating the primary 'equatorial' component and resulting in predominance of 'normal' directions, or a combination of these effects. There are not enough data to choose unequivocally any one of these alternatives and to reject the others and any other alternative explanations.

5.2 LATE MIOCENE

Opaque mineralogical studies of the Late Miocene lavas show that the stability of their primary remanence is high, and thermal 'cleaning' has effectively removed the secondary components. After application of local and regional tilt corrections, the final direction (Table 4) when reversed in sense gives $D = 342.6°$, $I = 38.5°$. This direction is significantly different from the direction of the present-day geomagnetic field and probably does not reflect a record of secular variation since field evidence showed that the sampling sites could be associated with three distinct extrusive events.

6 Palaeotectonic considerations

Although the data have been obtained from an unavoidably inadequate rock material, we consider their possible palaeomagnetic implications. The mean palaeomagnetic poles listed in Table 4 are displayed in Fig. 12. The 'normal' Late Cretaceous pole C at 143.8° W, 44.1° N

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corresponds to the mean of nine units, excluding group R(10), which is shown separately. Both observed and reversed 'equatorial' pole positions are shown for the mean of dykes S, T and V (E_1 and E_1') and for R(4). Western 'equatorial' pole positions are given for the means of six and seven 'equatorially' magnetized dykes (E_2 at 165.3°W, 5.6°N and E_3 at 165.7°W, 1.7°N respectively). The reversed pole position at 152.4°W, 73.3°N for the Low Layton lavas is shown. The mean positions of eight Cretaceous and six Miocene poles respectively for North American rocks, based on available data (Hicken *et al.* 1972), are also shown. The α_{95} values for these are 2.4 and 3.7° respectively.

In every case the 'clean' NRM directions are different from the present-day geomagnetic field direction and the palaeomagnetic poles have significantly different positions than those derived from the contemporaneous North American rocks.

Because of the scatter and ambiguity of the data, any palaeotectonic reconstructions must necessarily be highly speculative. Broadly speaking the data are consistent with the conclusion that Jamaica has undergone an anti-clockwise horizontal rotation since Late Cretaceous, a considerable portion of which occurred between the present and the Late Miocene. The rotation could have occurred *in situ* or about another rotational centre (LePichon, Francheteau & Bonnin 1973), producing an associated translation, or has been a combination of both these effects, all considered relative to the present-day frame of reference. An acceptable interpretation of the Miocene pole is a translation of Jamaica through about 1000 km from the southwest relative to the present-day frame of reference. The use of several palaeomagnetic poles for the determination of the rotational centre for a single tectonic plate has been considered by Cordell (1974), but in the present case only one Miocene pole is available. Moreover, the movement suggested by the Late Miocene data would have to be reconciled with that derived from 'equatorial' Cretaceous poles (E, Fig. 12) (assuming that the 'normal' poles reflect an incomplete removal of secondary remanence).

The ambiguity of interpretation of available data can be reduced by using two constraints. These are the westerly drift of North America since Late Mesozoic (LePichon *et al.* 1973; Solomon & Sleep 1974; Minster *et al.* 1974) and the present-day evidence for the relative easterly drift of the Caribbean (Molnar & Sykes 1969; Bracey & Vogt 1970; Isacks & Molnar 1971). On the basis of this evidence, the data are indeed consistent with a north-easterly displacement of Jamaica relative to the present-day reference frame and its additional anticlockwise rotation. This supports the suggestion that the Caribbean has behaved as an independent tectonic plate and that Jamaica, a smaller tectonic block on its northern margin, was capable of quasi-independent rotation.

The occurrence of 'normal' and of 'equatorial' directions of magnetization of opposite sense within the Effort group of dykes (Fig. 5) suggests that some of these dykes have been reheated after a polarity change of the geomagnetic field. It is not possible, however, to determine uniquely from the data whether this effect has, in fact, been instrumental in producing the 'equatorial' direction of magnetization or whether the latter was wholly or partly due to tectonic causes.

MacDonald & Opdyke (1972) interpret the results of palaeomagnetic studies of Cretaceous rocks on the Guajira Peninsula in Northern Colombia in terms of large scale horizontal rotations. A recent study of the magnetization of the Palaeocene Above Rocks pluton on Jamaica also suggests such rotations (Dasgupta & Vincenz 1975).

The total number of 'equatorial' pole positions derived from Jamaican rocks has been too great to allow an interpretation merely in terms of the behaviour of the geomagnetic field. A palaeotectonic interpretation in terms of a large rotation appears more appropriate. This view has recently been strengthened by preliminary palaeomagnetic studies of Hispaniola (Vincenz *et al.* 1973a; Vincenz & Dasgupta 1977).

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The palaeomagnetic data are clearly not capable of unique interpretation, but the above conclusions are consistent with evidence based on magnetometer surveys in the Colombian basin (Krause 1971; Christofferson 1973, 1976) and geophysical studies of the Cayman Trough (Ewing, Antoine & Ewing 1960; Bowin 1968; Gough & Heirtzler 1969; Holcombe *et al.* 1973; Heirtzler 1976; Ballard 1976) and of other features of Northern Caribbean (Burton 1965; Molnar & Sykes 1969; Bracey & Vogt 1970). According to this evidence the initially northerly sea-floor spreading was followed by an easterly left-lateral displacement of the Caribbean relative to the Atlantic plate, so that by Late Tertiary there was a relatively low accumulation of stresses at the Cayman Trough and a build-up of north–south compressive forces on the northeastern boundary of the Caribbean. This pattern of stress distribution is consistent with the proposed northeasterly movement of the Caribbean and its simultaneous anticlockwise rotation relative to North America.

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